BATMAN / LiFePO4 8S + lead-acid 24 V

Four PCBs: A ESP + two I2C ports, B analog filters, C 3V3 distribution, D 5V distribution.
Two external ADS1115: analog from B, power/I2C from A. B.Cu only; WC = external cables.
LOAD bus C1/C2 stay externally wired. Prototype: verify sensor model, fit and current budgets.

power

File: power.kicad_sch

charge

File: charge.kicad_sch

sense

File: sense.kicad_sch

logic

File: logic.kicad_sch

supply3

File: supply3.kicad_sch

supply5

File: supply5.kicad_sch

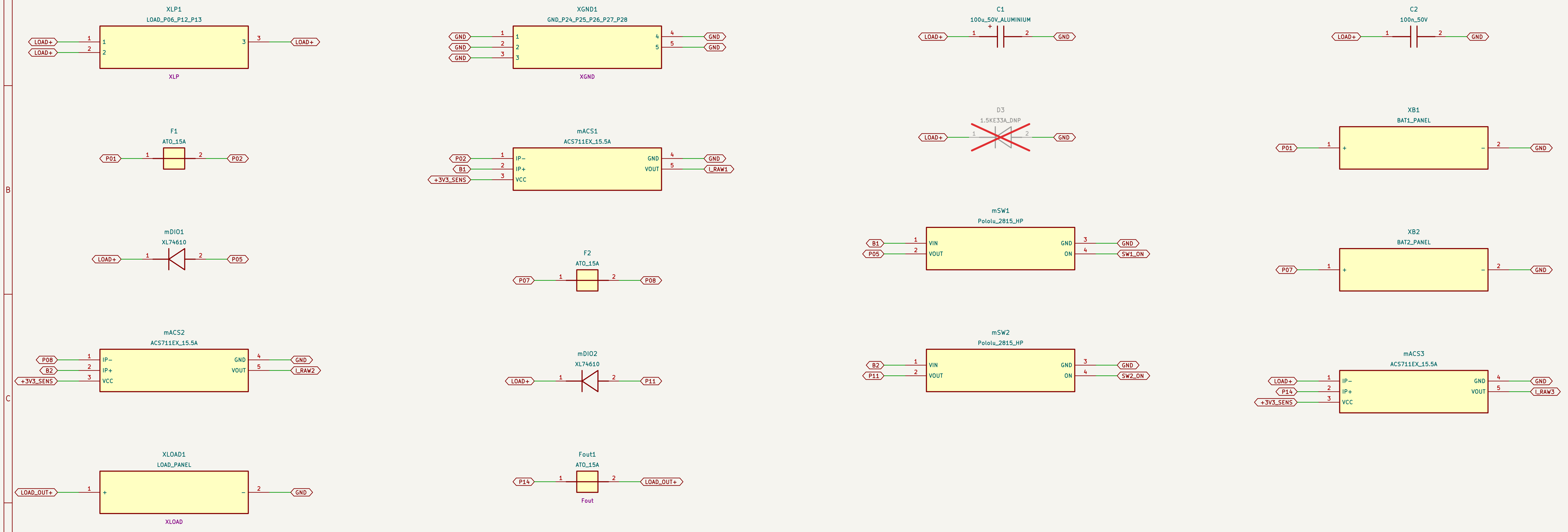
Sheet: /
File: batman.kicad_sch

Title: BATMAN – dual battery manager

Size: A4	Date: 2026-09-09	**Rev: A-prototype**
KiCad E.D.A. 10.0.6		Id: 1/7

BATMAN / POWER

P01–P28: external power wiring 2.5 mm2; P28 logic ground 0.75 mm2.
B1/B2 are AFTER sensors, BEFORE switches. IP– faces battery / source.
LOAD+/GND: external terminals, C1/C2 wired directly, no buffer PCB. D3 DNP; not a 35 V clamp.



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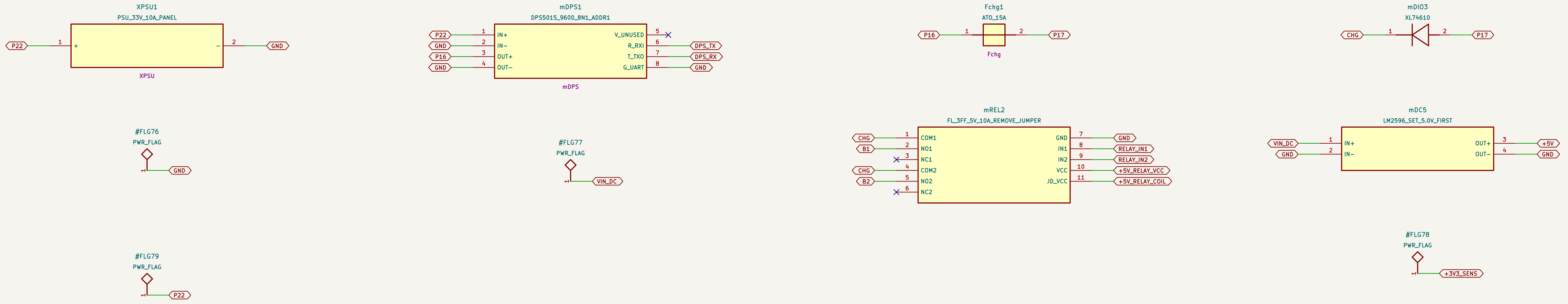
File: power.kicad_sch

Title: BATMAN – POWER

Size: A3 | Date: 2026-09-09 | **Rev: A-prototype**
KiCad E.D.A. 10.0.6 | Id: 2/7

BATMAN / CHARGE

REMOVE relay VCC–JD_VCC jumper. Route VCC / JD_VCC separately from X3.
DPS: GND TX RX –> G R T; leave V unconnected. Set mDC5 to 5.0 V BEFORE plugging J1.
Never close K1 and K2 together. Switch relays only at verified zero current. Hardware rating: 10 A maximum.



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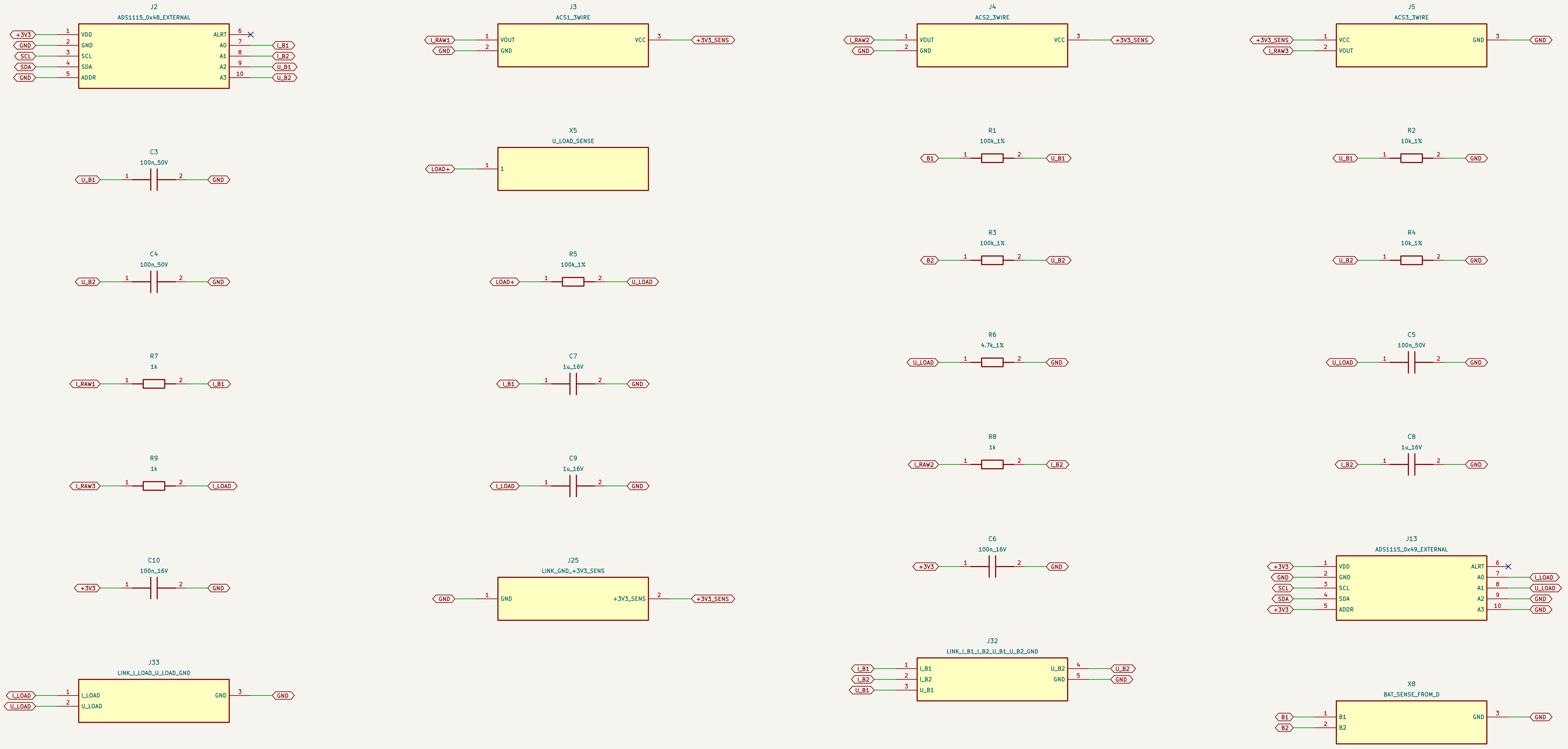
File: charge.kicad_sch

Title: BATMAN – CHARGE

Size: A3 | Date: 2026-09-09 | **Rev: A–prototype**
KiCad E.D.A. 10.0.6 | Id: 3/7

BATMAN / SENSE

PCB B: analog filters ONLY. J32 -> external J2 A0–A3/GND; J33 -> external J13 A0/A1/GND.
J2 ADDR=GND (0x48); J13 ADDR=VDD (0x49), A2/A3=GND; both ALRT NC. C6/C10 at modules.
ADCs powered from A, NOT B. J25: GND/3V3_SENS from C. No SDA/SCL on B. Verify sensor model.



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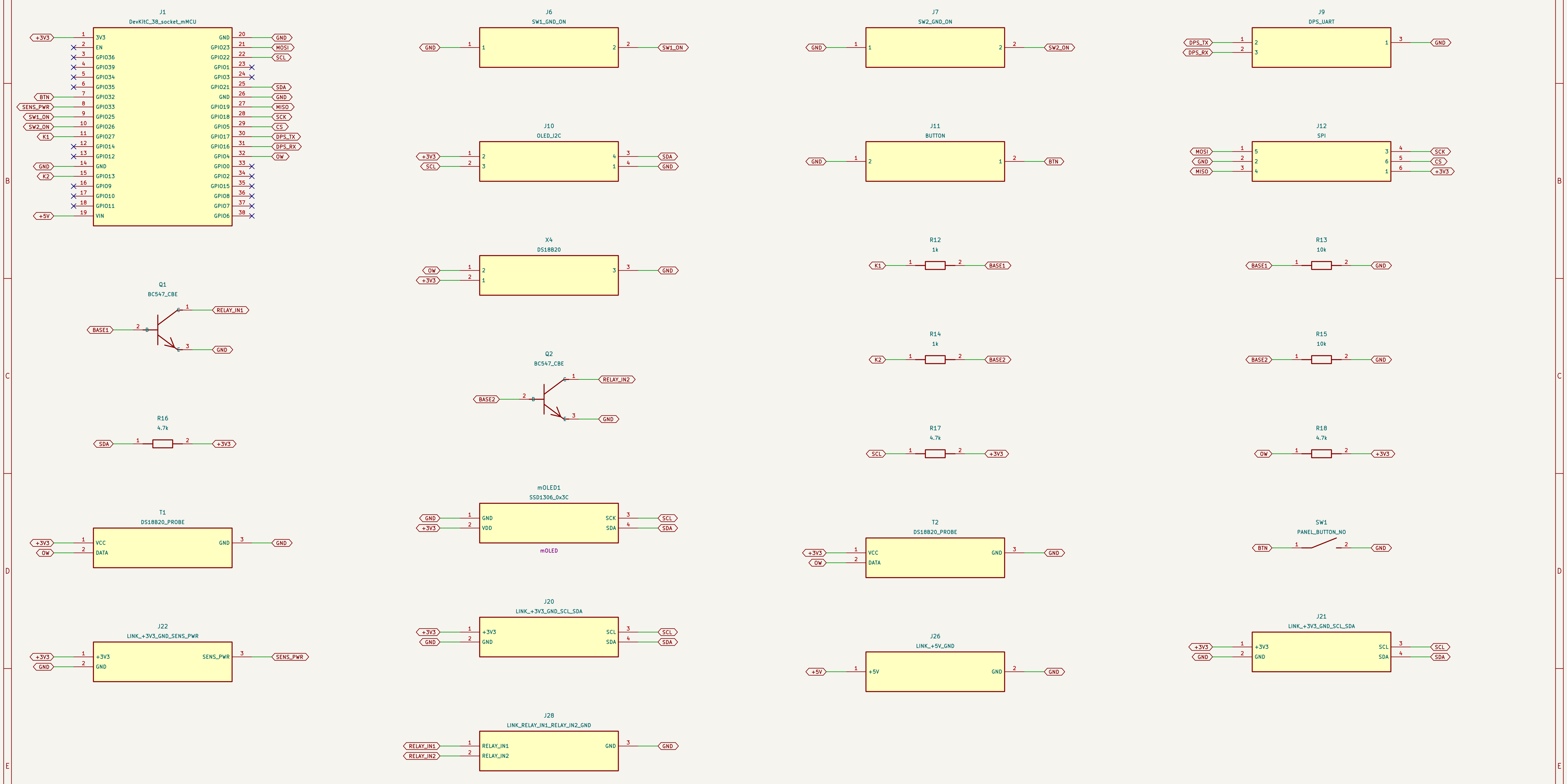
File: sense.kicad_sch

Title: BATMAN – SENSE

Size: A3 | Date: 2026-09-09 | **Rev: A-prototype**
KiCad E.D.A. 10.0.6 | Id: 4/7

BATMAN / LOGIC

PCB A: J20 -> external ADS J2; J21 -> external ADS J13. Both: 3V3/GND/SCL/SDA.
Shared GPIO21=SDA/GPIO22=SCL; one pull-up pair R16/R17. GPIO34/39 NC.
J22 -> C, J26 receives 5 V from D; J28 relay control to D. Cable map: output/cable-map.csv.



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File: logic.kicad_sch

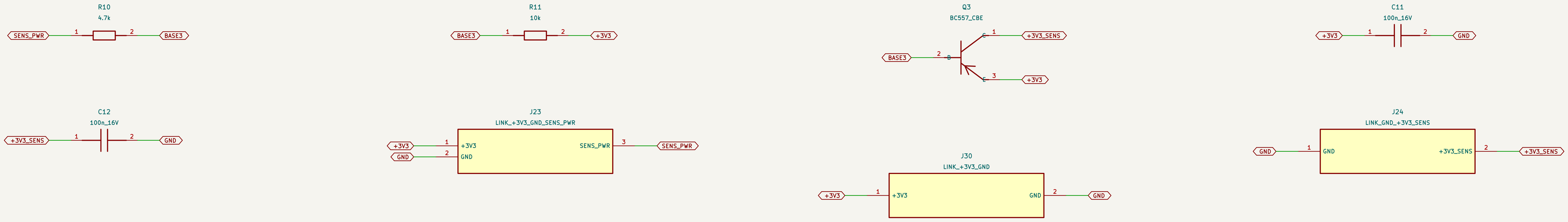
Title: BATMAN – LOGIC

Size: A3 | Date: 2026-09-09 | **Rev: A-prototype**

KiCad E.D.A. 10.0.6 | Id: 5/7

BATMAN / SUPPLY3

PCB C: DISTRIBUTION ONLY, no regulator. J23 receives 3V3/GND/SENS_PWR from ESP PCB A.
J24 supplies PCB B sensors: GND/3V3_SENS. J30 is a spare 3V3/GND output.
Q3 switches sensor supply only. Do NOT connect another regulator to this 3V3 rail.



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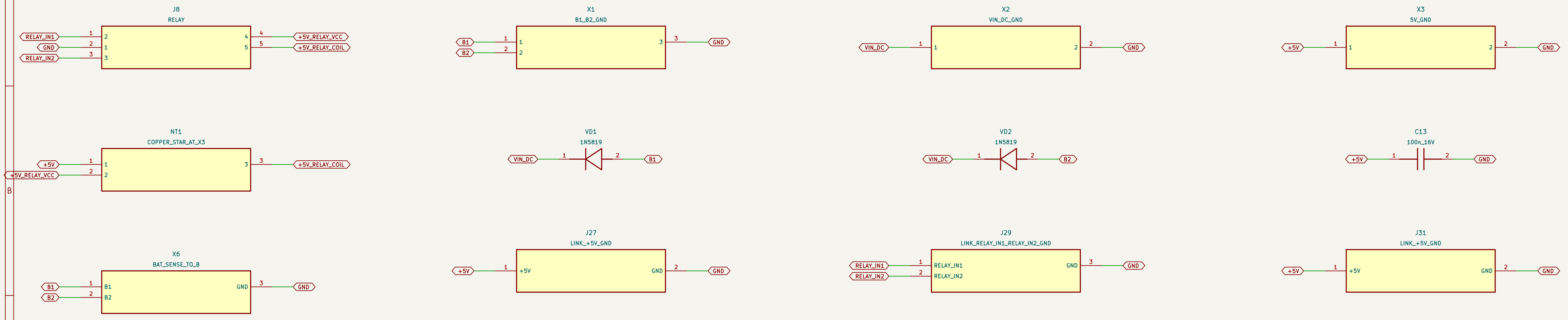
File: supply3.kicad_sch

Title: BATMAN – SUPPLY3

Size: A3	Date: 2026-09-09	**Rev: A-prototype**
KiCad E.D.A. 10.0.6		Id: 6/7

BATMAN / SUPPLY5

PCB D: DISTRIBUTION ONLY. X3 receives regulated 5.0 V from external LM2596.
X1 battery taps feed diode OR -> X2 LM2596 input; X6 -> X8 on ADC PCB B (sense only).
NT1 star: separate relay VCC/JD-VCC. REMOVE relay jumper. J27 -> ESP; J31 spare 5 V.



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Title: BATMAN – SUPPLY5

Size: A3	Date: 2026-09-09	Rev: A-prototype
KiCad E.D.A. 10.0.6		Id: 7/7